A COVERT BIDIRECTIONAL ACTIVE-MATRIX ILLUMINATION AND IMAGING APPARATUS

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Title: A COVERT BIDIRECTIONAL ACTIVE-MATRIX ILLUMINATION AND IMAGING APPARATUS

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 26714 Valleyside Drive, Katy, TX 77494, submitting this application while under a *de facto* office arrest at 2302 Britton Ridge Drive, Lake Pointe Estates, Katy, Texas 77494, created by a contested Void Capias Order on Case 25-DCV-328122 on the 387th District Court of Fort Bend County, Texas, currently under review on the Supreme Court of the United States Docket 26-A39 (published as a printed book with ISBN 9781971481616) being reviewed by Associate Justice Samuel A. Alito, Jr., and also under review under Petition for Writ of Habeas Corpus submitted to the Supreme Court of the United States on July 16th, 2026 (also published as printed book 9781971685700). The Inventor has submitted other Patent applications related to Post-Quantum Encryption, Optical Computing, National Defense, and related technologies which are published under ISBN 9781971685007 and 9781971481296.

Cross-Reference to Related Applications

This application is related to and incorporates by reference the entirety of U.S. Patent Application No. 19/746,749, titled "A MONOLITHIC BIDIRECTIONAL ACTIVE-MATRIX DISPLAY WITH ENDOGENOUS SELF-FEEDBACK IMAGING (P.E.R.W.I.N.)," filed on July 20, 2026 (Confirmation #5096, Patent Center #78998550).

Abstract

A covert illumination and imaging apparatus indistinguishable from a standard consumer light emitting diode (LED) bulb. The apparatus comprises a curved, flexible polymer substrate conforming to the interior geometry of a translucent bulb enclosure. The substrate supports an interleaved array of self-emissive subpixels (e.g., OLED or QDEL) and micro-scale organic photodetector (OPD) sensor subpixels. Driven by a de-phased duty cycle, the light-emitting subpixels provide room illumination matching the standardized lumen output of a conventional bulb. During sub-millisecond dark intervals, the integrated OPD sensors capture wide-angle, omnidirectional images of the surrounding environment. The system enables undetectable covert surveillance without the use of discrete, visually detectable camera modules or physical apertures.

Background of the Invention

Covert surveillance devices are frequently disguised within common household objects, such as smoke detectors, clock radios, and light bulbs. However, conventional covert light bulb cameras rely on traditional, discrete CMOS camera modules hidden behind a small physical aperture or a one-way mirrored section of the bulb's casing. These discrete camera lenses are highly vulnerable to detection by counter-surveillance sweeps using optical lens finders, thermal imaging (due to localized heat from the camera SoC), or simple physical inspection of the bulb's surface. Furthermore, the discrete camera occupies internal volume, reducing the available space for illumination hardware and leading to asymmetrical or suspicious light dispersion. There is a critical need for a covert imaging device that is physically and optically integrated into the light-emitting surface itself, rendering the camera absolutely undetectable by conventional counter-surveillance techniques.

Summary of the Invention

The present invention resolves the vulnerabilities of discrete covert cameras by applying a bidirectional active-matrix architecture (P.E.R.W.I.N.) to a three-dimensional, omnidirectional illumination form factor.

Rather than hiding a camera inside a bulb, the invention transforms the entire light-emitting surface of the bulb into the camera. The apparatus utilizes a flexible polymer substrate (such as polyimide) that is physically shaped or thermoformed to map to the interior surface of a

standard bulb enclosure (e.g., an A19 bulb, BR30 flush mount, or GU10 floodlight).

Deposited onto this curved substrate is an interleaved matrix of self-emissive subpixels (capable of outputting single-color white light, or variable RGB light) and Organic Photodetector (OPD) sensor subpixels.

The device is driven by a logic controller housed in the base of the bulb (e.g., within the E26 screw base). This controller executes a high-frequency, de-phased duty cycle. For example, operating at 120 Hz, the controller globally deactivates the emissive subpixels for a 500μs interval. During this precise dark window, the OPD sensors are activated to integrate the ambient light reflecting off the subjects in the room. Because the integration window is extremely short and the switching frequency is well above the human flicker fusion threshold, the bulb appears to emit a continuous, high-lumen output identical to a standard commercial LED bulb.

The curved nature of the substrate provides an inherent 360° or hemispherical field of view. The raw sensor data is processed by a computational deconvolution matrix to reconstruct a panoramic outward image, which is subsequently exfiltrated via integrated wireless communication (e.g., Wi-Fi, Bluetooth) or modulated directly over the AC power mains using Power Line Communication (PLC).

Claims

What is claimed is:

1. A covert illumination and imaging apparatus, comprising:
    - a translucent enclosure shaped as a standard illumination bulb;
    - a curved active-matrix substrate conforming to at least a portion of an interior surface of said enclosure;
    - an interleaved array of light-emitting subpixels and light-receiving photodetector subpixels fabricated on said curved active-matrix substrate; and
    - a driver circuit configured to execute a de-phased duty cycle, wherein said light-emitting subpixels are driven to emit visible illumination into a surrounding environment, and wherein said photodetector subpixels are selectively activated to capture ambient light during intermittent dark intervals when said light-emitting subpixels are deactivated.

2. The apparatus of claim 1, wherein the light-emitting subpixels comprise organic light-emitting diodes (OLED) or quantum dot electroluminescent (QDEL) diodes.

3. The apparatus of claim 1, wherein the photodetector subpixels comprise organic photodetectors (OPD).

4. The apparatus of claim 1, wherein the total light output of the light-emitting subpixels is configured to match the standardized lumen output of a conventional single-purpose light bulb.

5. The apparatus of claim 1, wherein the curved active-matrix substrate provides a panoramic or substantially hemispherical field of view for the photodetector subpixels.

6. The apparatus of claim 1, further comprising a computational processor housed within a base of the apparatus, said processor configured to reconstruct an image of the surrounding environment from voltage signals generated by the photodetector subpixels.